

Time Series Analysis

Chapter 3 — Stationary time series

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Chapter 1

Stationary time series

Introduction

After decomposing the observed series into trend, seasonality and residual (Chapter ??), we usually want the residual to be *stationary* so that the modelling tools of Chapter ?? apply. This chapter formalises stationarity, introduces the autocovariance and autocorrelation functions (ACF and PACF), states the Wold decomposition theorem, and provides *white-noise tests* (Box–Pierce, Ljung–Box) to validate model residuals.

1.1 Stationarity

1.1.1 Strict and weak stationarity

Definition 1 – Strict (strong) stationarity

A process $\{X_t\}_{t \in \mathbb{Z}}$ is *strictly stationary* if, for every integer k , every $t_1 < \dots < t_k$ and every $h \in \mathbb{Z}$,

$$(X_{t_1}, X_{t_2}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, X_{t_2+h}, \dots, X_{t_k+h}).$$

Definition 2 – Weak (covariance) stationarity

A process $\{X_t\}$ with $\mathbb{E}[X_t^2] < \infty$ is *weakly stationary* if there exist $\mu \in \mathbb{R}$ and a function $\gamma : \mathbb{Z} \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[X_t] = \mu, \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for all } t, h \in \mathbb{Z}.$$

The function γ is the *autocovariance function* (ACVF).

Proposition 3 – Strict $\Rightarrow$ weak (under L^2)

If $\{X_t\}$ is strictly stationary with $\mathbb{E}[X_t^2] < \infty$, then $\{X_t\}$ is weakly stationary.

Proof. By strict stationarity, $X_t \stackrel{d}{=} X_0$, so $\mathbb{E}[X_t] = \mathbb{E}[X_0] = \mu$ does not depend on t . Similarly, $(X_t, X_{t+h}) \stackrel{d}{=} (X_0, X_h)$, hence $\text{Cov}(X_t, X_{t+h}) = \text{Cov}(X_0, X_h)$ depends only on h . $\square$

Remark – The converse is false in general

A process can be weakly stationary without being strictly stationary if it has time-varying higher-order moments. For Gaussian processes (only mean and covariance matter), the

two notions coincide.

1.1.2 Two canonical examples

Definition 4 – White noise

(ε_t) is a *white noise* $\text{WN}(0, \sigma^2)$ if $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}[\varepsilon_t] = \sigma^2$ and $\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$ for $h \neq 0$. A *strong white noise* requires the ε_t to be i.i.d.; a Gaussian white noise has $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.

Example 5 – Random walk is *not* stationary

Let $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \text{WN}(0, \sigma^2)$ and $X_t = X_{t-1} + \varepsilon_t$ with $X_0 = 0$. Then $X_t = \sum_{s=1}^t \varepsilon_s$ and

$$\mathbb{E}[X_t] = 0, \quad \text{Var}[X_t] = t\sigma^2, \quad \text{Cov}(X_t, X_{t+h}) = t\sigma^2 \text{ for } h \geq 0.$$

The variance grows linearly in t , so X_t is non-stationary. This is the canonical example used by unit-root tests in Chapter ??.

1.2 Autocovariance and autocorrelation

Definition 6 – ACVF and ACF

For a weakly stationary process with mean μ :

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}), \quad \rho(h) = \frac{\gamma(h)}{\gamma(0)} \in [-1, 1].$$

Proposition 7 – Properties of γ

1. $\gamma(0) = \text{Var}[X_t] \geq 0$;
2. $|\gamma(h)| \leq \gamma(0)$ (Cauchy–Schwarz);
3. γ is even: $\gamma(-h) = \gamma(h)$;
4. γ is positive semidefinite: for any n and any $a_1, \dots, a_n \in \mathbb{R}$, $\sum_{i,j=1}^n a_i a_j \gamma(t_i - t_j) \geq 0$.

Proof. (1) and (3) follow from the definition. (2) follows from Cauchy–Schwarz applied to $X_t - \mu$ and $X_{t+h} - \mu$. (4) is the standard fact that any matrix of the form $\Sigma_{ij} = \gamma(t_i - t_j)$ is the covariance matrix of $(X_{t_1}, \dots, X_{t_n})$, hence positive semidefinite. $\square$

1.2.1 Sample ACF and Bartlett bands

Definition 8 – Sample ACVF and ACF

For $X_1, \dots, X_T$ with sample mean $\bar{X}$:

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (X_t - \bar{X})(X_{t+h} - \bar{X}), \quad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Theorem 9 – Bartlett’s formula for white noise

If $\{X_t\}$ is a white noise with finite fourth moment, then for any fixed $h \geq 1$:

$$\sqrt{T} \hat{\rho}(h) \xrightarrow{d} \mathcal{N}(0, 1).$$

The standard test bands $\pm 1.96/\sqrt{T}$ on the ACF plot are asymptotic 95% confidence intervals *under the white-noise null*.

Remark

Bartlett’s formula gives a more general result for non-white processes (involves a sum over the squared autocorrelations); see Brockwell & Davis, *Time Series: Theory and Methods*, Chapter 7.

1.3 Partial autocorrelation

Definition 10 – Partial autocorrelation function

The PACF $\alpha(h)$ at lag h is the correlation between X_t and X_{t+h} *after removing the linear effect* of the intermediate observations $X_{t+1}, \dots, X_{t+h-1}$:

$$\alpha(h) = \text{Corr}(X_t - \hat{X}_t, X_{t+h} - \hat{X}_{t+h}),$$

where $\hat{X}_t$ and $\hat{X}_{t+h}$ are the linear projections of X_t and X_{t+h} on the span of $\{X_{t+1}, \dots, X_{t+h-1}\}$.

Remark – Recursive computation

The PACF is computed by the *Durbin–Levinson* algorithm: it solves the Yule–Walker equations of order h and reads off the coefficient of $X_{t+h-h} = X_t$ in the linear regression of X_{t+h} on $X_{t+1}, \dots, X_{t+h}$. Most software (Python `statsmodels`, R `stats`) returns the PACF in $O(h)$ operations per lag.

1.4 ACF / PACF signatures of standard processes

Process	$\rho(h)$ behaviour	$\alpha(h)$ behaviour
WN	$\rho(h) = 0$ for $h \neq 0$	$\alpha(h) = 0$ for $h \geq 1$
AR(p)	decays (exp. / sinusoidal)	cuts off at $h = p$
MA(q)	cuts off at $h = q$	decays
ARMA(p, q)	decays after $h > q$	decays after $h > p$

Box–Jenkins identification. Plot the sample ACF and PACF, then use the table above to read off candidate orders (p, q) .

1.4.1 AR(1) and MA(1) — closed-form ACF

Example 11 – AR(1) ACF

For $X_t = \phi X_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ and $|\phi| < 1$:

$$\rho(h) = \phi^{|h|}, \quad h \in \mathbb{Z}, \quad \alpha(1) = \phi, \quad \alpha(h) = 0 \text{ for } h \geq 2.$$

The ACF decays exponentially; the PACF cuts off at lag 1.

Example 12 – MA(1) ACF

For $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$:

$$\rho(0) = 1, \quad \rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h \geq 2.$$

The ACF cuts off at lag 1; the PACF decays.

Proof of $\rho(1) = \theta/(1 + \theta^2)$ in MA(1). $\text{Var}[X_t] = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})^2] = (1 + \theta^2)\sigma^2$. Then $\text{Cov}(X_t, X_{t+1}) = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t+1} + \theta \varepsilon_t)] = \theta \sigma^2$ since ε_t 's are uncorrelated. Hence $\rho(1) = \theta \sigma^2 / [(1 + \theta^2)\sigma^2] = \theta/(1 + \theta^2)$. $\square$

1.5 Wold decomposition

Theorem 13 – Wold's theorem, 1938

Every *purely non-deterministic* weakly stationary process $\{X_t\}$ admits a unique representation

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1, \quad \sum_{j=0}^{\infty} \psi_j^2 < \infty,$$

where (ε_t) is a white-noise process (the *innovations*) of variance $\sigma^2 = \text{Var}[\varepsilon_t]$.

Remark – Why it matters

- **Universality:** every stationary process can be viewed as an *infinite MA process*.
- **Justification of ARMA:** ARMA models are finite-dimensional approximations of the Wold representation (with a rational $\psi(z)$).
- **Forecasting:** optimal linear predictors and their prediction errors are obtained from the Wold coefficients.

1.6 White-noise tests

After fitting a decomposition or an ARMA model, the residuals $\hat{\varepsilon}_t$ should look like white noise. Two classical tests check this jointly across several lags.

Definition 14 – Box–Pierce and Ljung–Box statistics

For sample ACF $\hat{\rho}(h)$ of the residuals up to lag m :

$$Q_{BP} = T \sum_{h=1}^m \hat{\rho}(h)^2, \quad Q_{LB} = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}(h)^2}{T-h}.$$

Theorem 15 – Asymptotic distribution under H_0

Under H_0 : *the residuals are white noise*, both statistics are asymptotically χ^2_{m-p-q} , where $p+q$ is the number of fitted ARMA parameters (zero if no parametric model has been fitted).

Decision rule. Reject H_0 at level α if $Q > \chi^2_{m-p-q, 1-\alpha}$, equivalently $p\text{-value} < \alpha$.

Choosing m . A few rules of thumb:

- For yearly data: $m \in \{10, 20\}$.
- For monthly data: $m \in \{12, 24, 36\}$ (covers seasonality).
- For high-frequency data: m proportional to expected autocorrelation length.

Warning

The Ljung–Box test is preferred over Box–Pierce in finite samples: its small-sample distribution is closer to the χ^2 asymptotic limit, especially when m/T is not negligible.

Summary**Key takeaways.**

- *Strict stationarity* = invariance of all finite-dimensional distributions; *weak stationarity* = only first two moments. Strict $\Rightarrow$ weak (under L^2); converse holds for Gaussian processes.
- The *autocovariance* $\gamma(h)$ describes the dependence structure; the *autocorrelation* $\rho(h)$ is its normalised version.
- ACF and PACF have characteristic patterns that allow *Box–Jenkins identification* of AR(p) / MA(q) / ARMA(p,q).
- *Wold decomposition*: every stationary process is an infinite moving average of innovations.
- *Ljung–Box test* validates that residuals look like white noise.